MEMO

Paper 2 mock: marking memo

Tick-per-mark answers, the reasons DBE memos accept, and the traps that cost marks in real memos.

Marking rules that apply to every question (from the DBE memos)

- **Consistent accuracy (CA):** an error carried from an earlier part is not penalised again. Marking stops at the second calculation error in the same item.
- **Answer only:** full marks only where the memo says so (mostly regression a, b and r from the calculator). Otherwise working is needed.
- **Rounding:** two decimals unless asked. Early rounding that changes the final answer loses the answer mark.
- **Geometry:** marks are for statement AND reason (S, R), or one mark for both (S/R). A correct statement with no reason usually earns nothing. Use the accepted reasons, e.g. *tan chord theorem*, *$\angle$s in the same seg*, *opp $\angle$s of cyclic quad*, *line $\parallel$ one side of Δ*. Never assume something you were not given.
- **Trig:** "without a calculator" means every step must be shown; a calculator value gets 0. A general solution needs $k \in \mathbb{Z}$.
- **Identities:** work on one side only; do not move terms across the = sign.

What is NOT on the information sheet (learn it)

Geometry reasons DBE accepts (write them like this)

- line from centre ⊥ to chord · line from centre to midpt of chord
- ∠ at centre = 2 × ∠ at circumference
- ∠ in semi circle · ∠s in the same seg · equal chords; equal ∠s
- opp ∠s of cyclic quad · ext ∠ of cyclic quad
- tan ⊥ radius · tans from common pt · tan chord theorem
- converse ∠s in the same seg · converse opp ∠s of cyclic quad · converse ext ∠ of cyclic quad · converse tan chord theorem · converse ∠ in semi circle
- line ∥ one side of △ (prop theorem; name the ∥ lines) · line divides two sides of △ in prop (converse) · midpt theorem
- ||| △s · ∠∠∠ (equiangular) · sides of △ in prop · SSS, SAS, AAS, RHS (≡)
- ∠s on a str line · vert opp ∠s · ∠s around a pt · corresp ∠s; AB ∥ CD · alt ∠s · co-int ∠s · sum of ∠s in △ · ext ∠ of △ · ∠s opp equal sides · sides opp equal ∠s · radii · Pythagoras

Only these proofs are examinable

- Line from centre ⊥ chord / to midpoint of chord (8.5.1)
- ∠ at centre = 2 × ∠ at circumference (8.5.2)
- Opp ∠s of cyclic quad supplementary (8.5.3)
- Tan chord theorem (8.5.4)
- Proportionality theorem (10.1)
- Equiangular △s have sides in proportion (10.2)

Trig you must know

- $\tan x = \dfrac{\sin x}{\cos x}$; $\sin^2 x + \cos^2 x = 1$
- CAST: All (Q1), Sin (Q2), Tan (Q3), Cos (Q4) are positive
- $180° \pm x$, $360° - x$, $-x$: same function, sign from CAST
- $90° \pm x$: sin ↔ cos, sign from CAST, e.g. $\cos(90° + x) = -\sin x$
- $\sin(x + k \cdot 360°) = \sin x$; $\tan(x + k \cdot 180°) = \tan x$
- Special angles: $\sin 30° = \cos 60° = \dfrac{1}{2}$, $\sin 60° = \cos 30° = \dfrac{\sqrt{3}}{2}$, $\sin 45° = \cos 45° = \dfrac{1}{\sqrt{2}}$, $\tan 30° = \dfrac{1}{\sqrt{3}}$, $\tan 45° = 1$, $\tan 60° = \sqrt{3}$
- General solution: $\sin x = c$: $x = \theta + k \cdot 360°$ or $180° - \theta + k \cdot 360°$; $\cos x = c$: $x = \pm\theta + k \cdot 360°$; $\tan x = c$: $x = \theta + k \cdot 180°$; $k \in \mathbb{Z}$
- Period: $\sin bx$, $\cos bx$: $\dfrac{360°}{b}$; $\tan bx$: $\dfrac{180°}{b}$

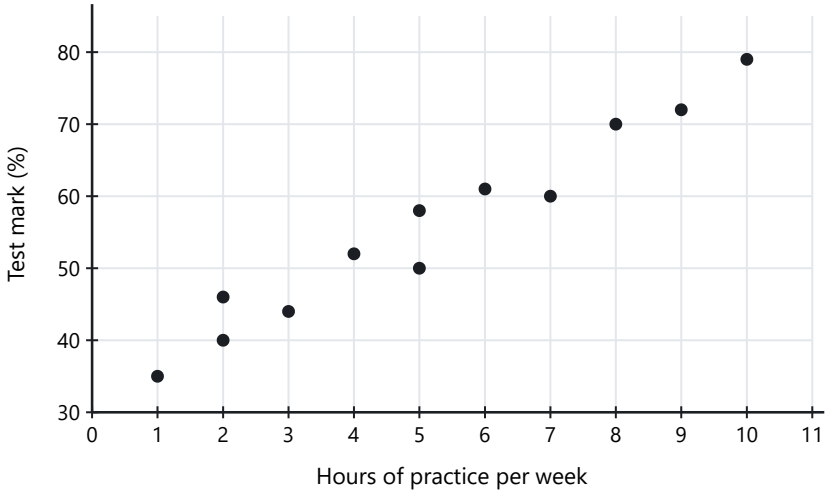
Analytical geometry and statistics

- ∥ lines: $m_1 = m_2$; ⊥ lines: $m_1 \times m_2 = -1$
- Angle between two lines = difference of their inclinations (ext ∠ of △)
- Tangent ⊥ radius at the point of contact; circles touch: distance between centres $= r_1 + r_2$ (outside) or $|r_1 - r_2|$ (inside)
- Calculator: STAT → A+BX for a, b, r ; 1-VAR for $\bar{x}$ and σx (population SD)
- Ogive: plot at UPPER class limits, start at (lowest limit ; 0)
- Adding a constant to every value: mean changes, SD does not

QUESTION 1

1.1 12 points plotted correctly, e.g. (1 ; 35), (5 ; 58), (10 ; 79) ✓✓✓ (3)

3 marks: all 12 correct; 2 marks: 8–11 correct; 1 mark: 4–7 correct.



1.2 Strong (positive) relationship ✓ (1)

The points lie close to a straight line that slopes upwards.

1.3 $a = 32{,}15$ ✓ (3)

$b = 4{,}54$ ✓

$\hat{y} = 32{,}15 + 4{,}54x$ ✓

Calculator (STAT, A+BX): $a = 32{,}1495\ldots$; $b = 4{,}5356\ldots$. Answer only: full marks.

1.4 $r = 0{,}98$ ✓ (1)

$r = 0{,}9774\ldots$

1.5 Very strong positive correlation ✓ (1)

Memo also accepts 'strong positive'. The word *positive* or the direction must be there.

1.6 $\hat{y} = 32{,}15 + 4{,}54(6{,}5)$ ✓ (2)

$= 61{,}66\,\%$ ✓

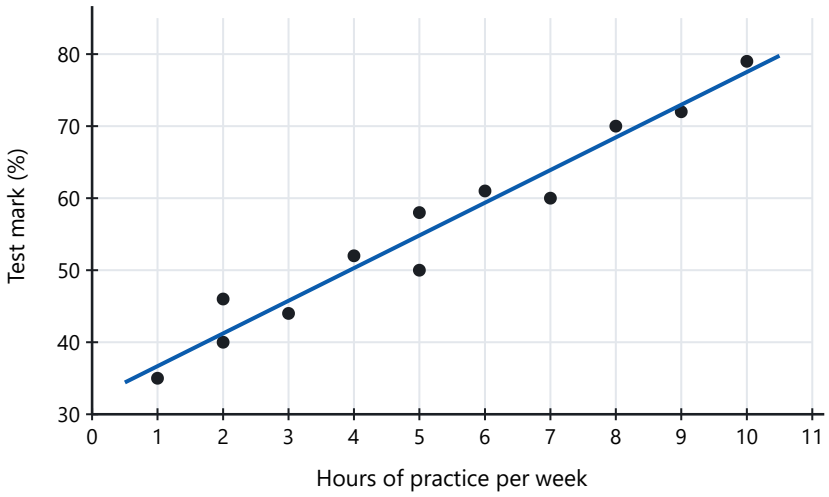
Using the calculator values directly: $61{,}63\,\%$. Both are accepted.

1.7 $r = 0{,}98$ is very close to 1 ✓: the points lie very close to the line, so the prediction is reliable. (2)

✓

1.8 Straight line through any two correct points, e.g. $(0; 32{,}15)$ and $(10; 77{,}55)$, or through $(\bar{x}; \bar{y}) = (5{,}17; 55{,}58)$ ✓✓ (2)

The line must pass through $(\bar{x}; \bar{y})$.



1.9 About $4{,}54$ percentage points per extra hour (the gradient b) ✓ (1)

1.10 Decrease ✓ – the point (11 ; 48) lies far below the line at the right-hand end, so it pulls that end of the line down. ✓ (2)

1.11 C ✓✓ (2)

At $x = 9$ the line predicts about $73\,\%$.

1.12 $\bar{y} = \dfrac{667}{12} = 55{,}58$ ✓✓ (3)

$\sigma = 12{,}96$ ✓

1.13 $\bar{y} - \sigma = 55{,}58 - 12{,}96 = 42{,}62$ ✓ (2)

Marks below 42,62: 35; 40, so 2 learners ✓

1.14 The point furthest BELOW the line is $(5; 50)$; the line gives $54{,}83$ there ✓ (2)

Increase $= 54{,}83 - 50 = 4{,}83$ marks ✓

In whole marks the learner needs 5 more marks; both answers are accepted.

1.15 C ✓ (1)

(Actual value: $r = 0{,}23$.)

1.16 The points are widely scattered about the line (weak), and the line slopes slightly upwards (positive). ✓ (1)

1.17 A shoe size of 0 does not exist, so no person can have $x = 0$ and a height of 82,4 cm. ✓ (1)

Same idea as the 2018 memo: a height of 0 m is impossible.

[30]

QUESTION 2

Data Handling: One Variable

2.1.1 $\bar{x} = \dfrac{181}{15}$ ✓$= 12{,}07$ minutes ✓ (2)

2.1.2 $\sigma = 6{,}42$ ✓ (1)

Use the calculator (population SD, σx). Answer only: full marks.

2.1.3 $(\bar{x} - \sigma; \bar{x} + \sigma) = (5{,}64; 18{,}49)$ ✓ (3)

Values inside: 6 ; 7 ; 7 ; 8 ; 9 ; 10 ; 10 ; 11 ; 12 ; 13 ; 15 ; 18 ✓

12 waiting times ✓

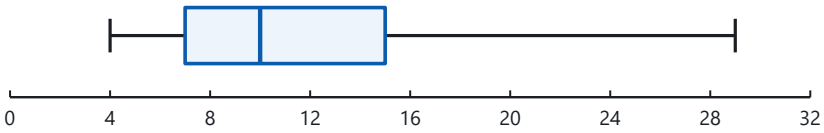
2.1.4 $(4; 7; 10; 15; 29)$ ✓✓✓ (3)

min ; Q_1 ; median ; Q_3 ; max. With 15 values the median is the 8th value; Q_1 and Q_3 are the 4th and 12th values (medians of the lower and upper 7).

2.1.5 $IQR = Q_3 - Q_1 = 15 - 7$ ✓$= 8$ ✓ (2)

2.1.6 $29 - 4 = 25$ minutes ✓ (1)

2.1.7 Whiskers from 4 to 29 ✓; box from 7 to 15 ✓; median at 10 ✓ (3)



2.1.8 Positively skewed (skewed to the right) ✓ (1)

The right whisker is much longer, and the mean (12,07) is bigger than the median (10).

2.1.9 Diagram Q ✓ – the median is 10 (or Q_1 = 7, or Q_3 = 15). ✓ (2)

2.2.1 The first rank (median 10 < 14) ✓ (1)

2.2.2 None ✓ – the shortest time at the second rank is 9 minutes, and 5 taxis at the first rank waited less than 9 minutes. ✓ (2)

2.2.3 (a) The totals are equal (same mean, same number of taxis) ✓ (2)
(b) The first rank's times vary more (σ = 6,42 > 2,1); the third rank's times are more consistent. ✓

2.2.4 (a) The mean increases by 1 minute ✓ (2)
(b) The standard deviation stays the same ✓

Adding the same number to every value moves the data but does not change its spread.

2.2.5 Mean $= 1{,}1 \times 12{,}07 = 13{,}27$ ✓; SD $= 1{,}1 \times 6{,}42 = 7{,}07$ ✓ (2)

Multiplying every value by a number multiplies the mean AND the standard deviation by it. Adding a number changes only the mean.

2.2.6 Monday total $= 181$; Tuesday total $= 15 \times 9 = 135$ ✓ (2)
Difference $= 46$ minutes ✓

2.2.7 $\dfrac{94 + x}{9} = 12 \Rightarrow x = 14$ ✓✓ (5)
$\sigma = 5{,}21$ ✓
Interval $(6{,}79; 17{,}21)$ ✓
Outside: 3 ; 18 ; 21 → 3 taxis ✓

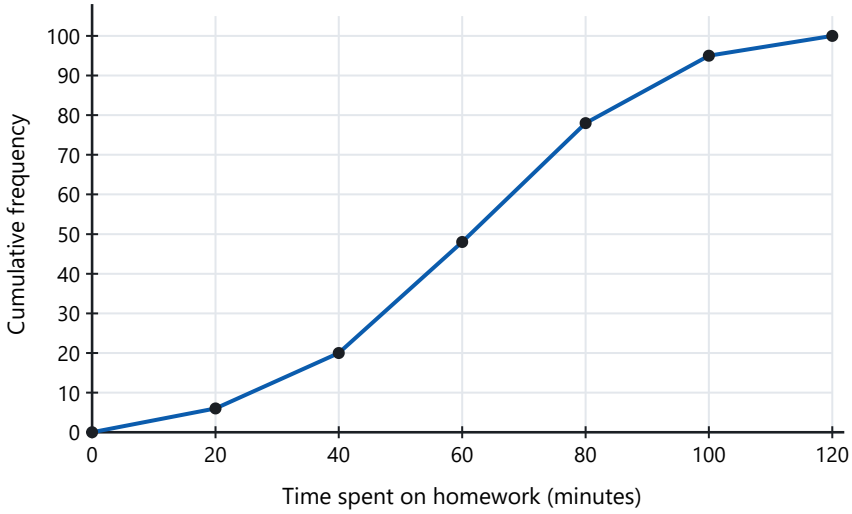
2.3.1 $6 + 14 + a + 30 + b + 5 = 100 \Rightarrow a + b = 45$ ✓ (5)
$\dfrac{10(6) + 30(14) + 50a + 70(30) + 90b + 110(5)}{100} = 60{,}6$ ✓✓
$3\,130 + 50a + 90b = 6060$ and $a = 45 - b$: $3\,130 + 2\,250 - 50b + 90b = 6060$ ✓
$40b = 680 \Rightarrow b = 17$ and $a = 28$ ✓

2.3.2 $60 < x \leq 80$ ✓ (1)

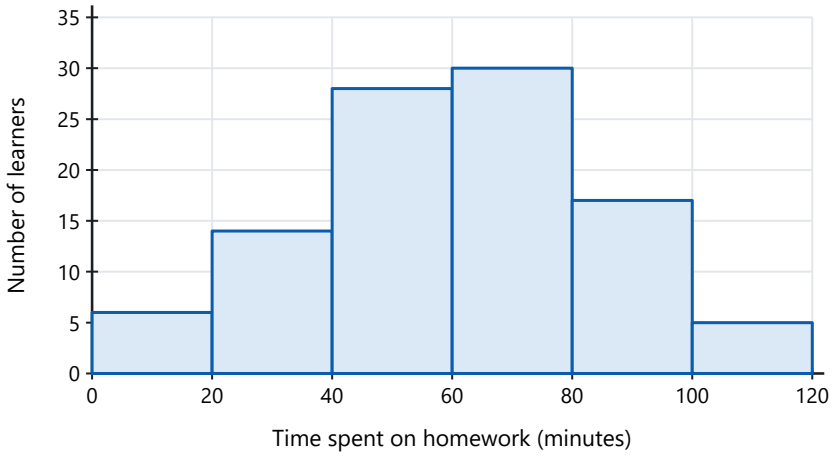
2.3.3 Cumulative frequencies: 6 ; 20 ; 48 ; 78 ; 95 ; 100 ✓✓ (2)

2.3.4 Grounded at (0 ; 0) ✓, plotted at the UPPER limits ✓, correct cumulative values ✓, joined ✓ (4)

Points: (0 ; 0) ; (20 ; 6) ; (40 ; 20) ; (60 ; 48) ; (80 ; 78) ; (100 ; 95) ; (120 ; 100)



2.3.5 Bars of heights 6 ; 14 ; 28 ; 30 ; 17 ; 5 on the six class intervals ✓✓, no gaps between the bars ✓ (3)



2.4.1 About 61 minutes ✓ (1)

Read at cumulative frequency 50. Accept 59–63.

2.4.2 100 learners ✓; $60 < x \leq 80$ (steepest part of the ogive) ✓ (2)

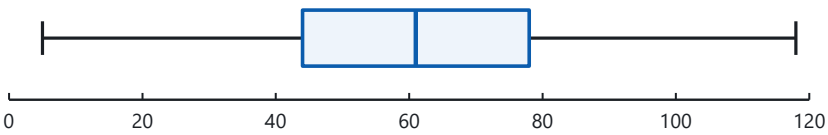
2.4.3 About 44 minutes ✓ (1)

Read at 25 (a quarter of 100).

2.4.4 $Q_3 \approx 78$ ✓; $IQR \approx 78 - 44 = 34$ minutes ✓ (2)

2.4.5 At 70 minutes the cumulative frequency is about 63 ✓; $100 - 63 = 37$ learners ✓ (2)

2.4.6 Box from 44 to 78 with median 61 ✓; whiskers to 5 and 118 ✓ (2)



2.4.7 The whole ogive shifts 10 units to the right ✓✓ (2)

Same shape, same height (still 100 learners).

2.4.8 Sum of (midpoint × frequency) before $= 6060$ ✓ (4)

$$\frac{6060 + \frac{k}{2}(10) + \frac{k}{2}(50)}{100 + k} = 55{,}5 \checkmark$$

$$6060 + 30k = 5550 + 55{,}5k \checkmark$$

$$k = 20 \checkmark$$

2.5.1 Monday (difference of 15) ✓ (1)

2.5.2 A: $\frac{275}{5} = 55$ ✓; B: $\frac{282}{5} = 56{,}4$ ✓ (2)

2.5.3 A ✓ (Mon, Wed, Thu). C is false: $\sigma_A = 11{,}90 > \sigma_B = 8{,}82$. ✓ (2)

[70]

QUESTION 3

Analytical Geometry: Straight Lines

3.1 $AB = \sqrt{(-2+5)^2 + (6-0)^2}$ ✓$= \sqrt{45} = 3\sqrt{5}$ ✓ (2)

3.2 $m_{\text{BC}} = \frac{3-0}{4-(-5)}$ ✓$= \frac{1}{3}$ ✓ (2)

3.3 $\tan\theta = \frac{1}{3}$ ✓$\theta = 18{,}43°$ ✓ (2)

3.4 $M\left(\frac{-5+4}{2}; \frac{0+3}{2}\right)$ ✓$= M\left(-\frac{1}{2}; \frac{3}{2}\right)$ ✓ (2)

3.5 $y - 0 = \frac{1}{3}(x+5)$ ✓✓ (3)

$y = \frac{1}{3}x + \frac{5}{3}$ ✓

3.6 $5y = -3x + 6 \Rightarrow y = -\frac{3}{5}x + \frac{6}{5}$ ✓ (3)

$m = -\frac{3}{5}$ ✓

$3\left(-\frac{1}{2}\right) + 5\left(\frac{3}{2}\right) = 6$ ✓, so M lies on the line

3.7 $m_{\text{AB}} = \frac{6-0}{-2+5} = 2$ and $m_{\text{AC}} = \frac{3-6}{4+2} = -\frac{1}{2}$ ✓ (2)

$m_{\text{AB}} \times m_{\text{AC}} = 2 \times \left(-\frac{1}{2}\right) = -1$, so AB ⊥ AC ✓

3.8 $\tan\alpha = m_{\text{AB}} = 2 \Rightarrow$ inclination of AB $= 63{,}43°$ ✓ (4)

$\text{A}\hat{\text{B}}\text{C} =$ inclination of AB − inclination of BC ✓

$= 63{,}43° - 18{,}43°$ ✓$= 45°$ ✓

Asked as 'calculate the size of Â' (2021) or 'LN̂S' (2023): always inclination of one line minus the other, using the exterior angle of a triangle.

3.9 $m = \frac{1}{3}$ [∥ lines] ✓ (3)

$6 = \frac{1}{3}(-2) + c$ ✓

$y = \frac{1}{3}x + \frac{20}{3}$ ✓

3.10 $AC = \sqrt{(4+2)^2 + (3-6)^2} = \sqrt{45}$ ✓ (3)

Â = 90°, so area $= \frac{1}{2}(AB)(AC) = \frac{1}{2}(\sqrt{45})(\sqrt{45})$ ✓

$= 22{,}5$ square units ✓

3.11 Area $= \frac{1}{2}(AB)(BC)\sin 45° = \frac{1}{2}(\sqrt{45})(\sqrt{90})\left(\frac{\sqrt{2}}{2}\right)$ ✓$= 22{,}5$ ✓ (2)

3.12 $DE = 5 - \frac{5}{3} = \frac{10}{3}$; height from C to the y-axis is 4 ✓ (4)

Area △CDE $= \frac{1}{2}\left(\frac{10}{3}\right)(4) = \frac{20}{3}$ ✓

Area ABDE = Area △ABC − Area △CDE $= 22{,}5 - \frac{20}{3}$ ✓

$= 15{,}83$ square units ✓

3.13 Â = 90°, so BC is a diameter of the circle through A, B and C [converse ∠ in semicircle] ✓ (3)

The centre is the midpoint of BC ✓

$\left(-\frac{1}{2}; \frac{3}{2}\right)$ ✓

3.14 $N\left(\frac{-2-5}{2}; \frac{6+0}{2}\right) = N\left(-\frac{7}{2}; 3\right)$ ✓ (4)

$m_{NM} = \frac{\frac{3}{2} - 3}{-\frac{1}{2} + \frac{7}{2}} = \frac{-\frac{3}{2}}{3} = -\frac{1}{2}$ ✓

$m_{AC} = -\frac{1}{2}$ ✓

$m_{NM} = m_{AC}$, so NM ∥ AC ✓

Or with geometry: N and M are midpoints of AB and BC [midpoint theorem].

3.15 Diagonals of a square bisect each other: midpoint of AK = midpoint of BC = M ✓ (3)

$\frac{-2 + x_K}{2} = -\frac{1}{2} \Rightarrow x_K = 1$ ✓

$\frac{6 + y_K}{2} = \frac{3}{2} \Rightarrow y_K = -3$; K(1 ; −3) ✓

Or by translation: A → B is (x − 3 ; y − 6), so C → K: (4 − 3 ; 3 − 6).

3.16 (5)

F lies on AC: $y = -\frac{1}{2}x + 5$ ✓

$(x+5)^2 + \left(-\frac{1}{2}x + 5\right)^2 = 65$ ✓

$\frac{5}{4}x^2 + 5x - 15 = 0 \Rightarrow x^2 + 4x - 12 = 0$ ✓

$(x+6)(x-2) = 0$, $x = 2$ (F between A and C, so $-2 < x < 4$) ✓

F(2 ; 4) ✓

3.17 $m_{CG} = \frac{6-3}{p-4} = \frac{3}{p-4}$ ✓ (2)

$\frac{3}{p-4} = -3 \Rightarrow p = 3$ ✓

3.18 (5)

$m_{AT} = -3$; AT: $y - 6 = -3(x + 2) \Rightarrow y = -3x$ ✓

$-3x = \frac{1}{3}x + \frac{5}{3} \Rightarrow x = -\frac{1}{2}$ ✓

$T\left(-\frac{1}{2}; \frac{3}{2}\right)$ ✓

$AT = \sqrt{\left(-\frac{1}{2} + 2\right)^2 + \left(\frac{3}{2} - 6\right)^2} = \sqrt{22{,}5} = 4{,}74$ ✓✓

T turns out to be M: in an isosceles triangle the perpendicular from the apex bisects the base.

3.19 (3)

P(7 ; 9) (not needed for the area)

Area = base × perpendicular height $= BC \times AT = \sqrt{90} \times \sqrt{22{,}5}$ ✓✓

$= 45$ square units ✓

3.20 (3)

$\frac{AH}{AC} = \frac{AG}{AB} = \frac{1}{3}$ [line ∥ one side of △] ✓

$\frac{\frac{1}{2}(AG)(AH)\sin A}{\frac{1}{2}(AB)(AC)\sin A}$ ✓

$= \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$ ✓

3.21 (3)

The angle between EC and the y-axis (below E) is $\mathrm{D\hat{E}C} = 63{,}43°$ ✓

By symmetry EC′ makes the same angle on the other side of the y-axis ✓

$\mathrm{C\hat{E}C'} = 2 \times 63{,}43° = 126{,}87°$ ✓

[63]

QUESTION 4

Analytical Geometry: The Circle

4.1 (3)

$x^2 - 4x + 4 + y^2 + 2y + 1 = 20 + 4 + 1$ ✓

$(x - 2)^2 + (y + 1)^2 = 25$ ✓

M(2 ; −1) and $r = 5$ ✓

4.2 (3)

$r^2 = (5 - 2)^2 + (3 + 1)^2 = 25$ ✓

$(x - 2)^2 + (y + 1)^2 = 25$ ✓

$x^2 - 4x + 4 + y^2 + 2y + 1 - 25 = 0 \Rightarrow x^2 + y^2 - 4x + 2y - 20 = 0$ ✓

4.3 (2)

M is the midpoint of PQ: $\frac{5 + x_Q}{2} = 2$ and $\frac{3 + y_Q}{2} = -1$ ✓

Q(−1 ; −5) ✓

4.4 (4)

$m_{MP} = \frac{3 - (-1)}{5 - 2} = \frac{4}{3}$ ✓

$m_{ST} = -\frac{3}{4}$ [radius ⊥ tangent] ✓

$3 = -\frac{3}{4}(5) + c$ ✓

$y = -\frac{3}{4}x + \frac{27}{4}$ ✓

4.5 (3)

$(7 - 2)^2 + (y + 1)^2 = 25 \Rightarrow (y + 1)^2 = 0$ ✓

only one solution, $y = -1$, so the line touches the circle ✓

point of contact (7 ; −1) ✓

4.6 The two tangents with this gradient have y-intercepts $\frac{27}{4}$ and $-\frac{23}{4}$ ✓ (3)

$-\frac{23}{4} < k < \frac{27}{4}$ ✓✓

Tangent at Q (∥ ST): $y = -\frac{3}{4}x - \frac{23}{4}$. NOT touch or cut: $k < -\frac{23}{4}$ or $k > \frac{27}{4}$. Touch: $k = -\frac{23}{4}$ or $\frac{27}{4}$.

4.7 $SM = \sqrt{(9-2)^2 + (0+1)^2} = \sqrt{50} > 5$, so S is outside ✓ (3)

$SP^2 = SM^2 - MP^2$ [radius ⊥ tangent; Pythagoras] $= 50 - 25$ ✓

$SP = 5$ ✓

4.8 Tangents from a common point are equal. ✓ (1)

4.9 In △MPS: $\tan P\hat{S}M = \frac{MP}{SP} = \frac{5}{5} = 1$ ✓✓ (4)

$\text{P}\hat{\text{S}}\text{M} = 45°$ ✓

$\text{P}\hat{\text{S}}\text{U} = 2 \times 45° = 90°$ [SM bisects the angle of kite MPSU] ✓

4.10 Area $= 2 \times \frac{1}{2}(MP)(SP) = 2 \times \frac{1}{2}(5)(5)$ ✓$= 25$ square units ✓ (2)

4.11 $(d-2)^2 + (3+1)^2 = 25 \Rightarrow (d-2)^2 = 9$ ✓ (2)

$d = 5$ or $d = -1$; so $d = -1$ ✓

4.12 $(x-2)^2 + (2x-10+1)^2 = 25$ ✓ (4)

$5x^2 - 40x + 60 = 0 \Rightarrow x^2 - 8x + 12 = 0$ ✓

$x = 2$ or $x = 6$ ✓

(2 ; −6) and (6 ; 2) ✓

4.13 $AB^2 = AM^2 - MB^2 = (t-2)^2 + (t+9)^2 - 25$ ✓ (2)

$= t^2 - 4t + 4 + t^2 + 18t + 81 - 25 = 2t^2 + 14t + 60$ ✓

4.14 $2t^2 + 14t + 60$ is a minimum at $t = -\frac{b}{2a} = -\frac{14}{4} = -\frac{7}{2}$ ✓ (3)

$AB = \sqrt{2\left(-\frac{7}{2}\right)^2 + 14\left(-\frac{7}{2}\right) + 60} = \sqrt{35{,}5}$ ✓

$= 5{,}96$ units ✓

4.15 $MN = 11 - (-1) = 12$ ✓ (4)

Touch externally: $p + 5 = 12 \Rightarrow p = 7$ ✓

Touch internally (M inside N's circle): $p - 5 = 12 \Rightarrow p = 17$ ✓✓

4.16 Internal touching: distance between centres $= 5 - 2 = 3$ ✓ (4)

New centre $(-10 + k; -1)$: $|2 - (-10 + k)| = 3$ ✓

$12 - k = 3$ or $12 - k = -3$ ✓

$k = 9$ or $k = 15$ ✓

4.17 New centre (−6 ; −1): $6 - 5 = 1$ unit ✓ (1)

4.18 Image of M(2 ; −1) is (−1 ; 2) ✓ (3)

distance $= \sqrt{(2+1)^2 + (-1-2)^2}$ ✓

$= \sqrt{18} = 3\sqrt{2}$ ✓

4.19 The common chord is bisected by the line of centres (equal circles), so the midpoint is the midpoint of the centres ✓: $\left(\frac{1}{2}; \frac{1}{2}\right)$ ✓ (2)

4.20 $r = 3$ (distance from K to the x-axis) ✓ ; $(x+4)^2 + (y-3)^2 = 9$ ✓ (2)

4.21 KD $= q$ and radius KA $=$ KE $= q + 2$ ✓ (4)

$(q+2)^2 = (q+1)^2 + q^2$ [Pythagoras] ✓

$q^2 - 2q - 3 = 0 \Rightarrow (q-3)(q+1) = 0$ ✓

$q = 3$ $(q > 0)$ ✓

[59]

QUESTION 5

Trigonometry: Values, Reduction, Identities And Equations

5.1.1 $k^2 = 17^2 - 15^2 = 64$ ✓; $k = -8$ (P is in the third quadrant) ✓ (2)

5.1.2 $\tan\theta = \dfrac{-15}{-8} = \dfrac{15}{8}$ ✓ (1)

5.1.3 $= -\cos\theta$ ✓$= -\left(-\dfrac{8}{17}\right) = \dfrac{8}{17}$ ✓ (2)

5.1.4 $\sin 2\theta = 2\sin\theta\cos\theta$ ✓$= 2\left(-\dfrac{15}{17}\right)\left(-\dfrac{8}{17}\right)$ ✓$= \dfrac{240}{289}$ ✓ (3)

Same method for $\cos 2\theta = 2\cos^2\theta - 1 = -\dfrac{161}{289}$.

5.1.5 $= \sin\theta\cos 60° + \cos\theta\sin 60°$ ✓ (4)

$= \left(-\dfrac{15}{17}\right)\left(\dfrac{1}{2}\right) + \left(-\dfrac{8}{17}\right)\left(\dfrac{\sqrt{3}}{2}\right)$ ✓✓

$= \dfrac{-15 - 8\sqrt{3}}{34}$ ✓

5.1.6 β is in the 2nd quadrant: $\cos\beta = -\dfrac{3}{5}$ ✓ (4)

$\cos(\theta - \beta) = \cos\theta\cos\beta + \sin\theta\sin\beta$ ✓

$= \left(-\dfrac{8}{17}\right)\left(-\dfrac{3}{5}\right) + \left(-\dfrac{15}{17}\right)\left(\dfrac{4}{5}\right)$ ✓

$= -\dfrac{36}{85}$ ✓

5.1.7 $\sin(\theta - 90°) = \dfrac{b}{34}$ ✓ (4)

$\sin(\theta - 90°) = -\sin(90° - \theta) = -\cos\theta$ ✓

$= \dfrac{8}{17}$ ✓

$b = 34 \times \dfrac{8}{17} = 16$ ✓

5.1.8 $\theta = 180° + 61{,}93° = 241{,}93°$ ✓ (3)

$\tan(2 \times 241{,}93° - 40°)$ ✓

$= 9{,}3$ ✓

5.2.1 $= \sqrt{1 - \cos^2 32°}$ ✓$= \sqrt{1 - t^2}$ ✓ (2)

Or draw a right-angled triangle: adjacent t, hypotenuse 1, opposite $\sqrt{1-t^2}$.

5.2.2 $= 2\cos^2 32° - 1$ ✓$= 2t^2 - 1$ ✓ (2)

5.2.3 $= 2\cos^2 16°$ ✓$= 1 + \cos 32°$ ✓$= 1 + t$ ✓ (3)

$\cos 2A = 2\cos^2 A - 1 \Rightarrow 2\cos^2 16° = 1 + \cos 32°$.

5.2.4 (4)

$= \sin(32^\circ - 30^\circ) = \sin 32^\circ \cos 30^\circ - \cos 32^\circ \sin 30^\circ$ ✓$= \sqrt{1-t^2} \cdot \dfrac{\sqrt{3}}{2} - t \cdot \dfrac{1}{2}$ ✓✓

$= \dfrac{\sqrt{3(1-t^2)} - t}{2}$ ✓

5.2.5 (2)

$\cos 55^\circ = \sin 35^\circ$ ✓$= \dfrac{k}{\sqrt{1+k^2}}$ ✓

Triangle: opposite k, adjacent 1, hypotenuse $\sqrt{1+k^2}$.

5.3 (3)

$\cos x = -\dfrac{3}{5}$, so x is in the THIRD quadrant: $x = -3$, $r = 5$, $y = -4$ ✓

$\tan x = \dfrac{-4}{-3} = \dfrac{4}{3}$ ✓

$\sin(360^\circ + x) = \sin x = -\dfrac{4}{5}$ ✓

5.4.1 (3)

$2\cos^2 \beta - 1 = \dfrac{7}{25}$ ✓$\cos^2 \beta = \dfrac{16}{25}$ ✓$\cos \beta = \dfrac{4}{5}$ ✓

5.4.2 (2)

$\sin^2 \beta = 1 - \dfrac{16}{25}$ ✓$\sin \beta = \dfrac{3}{5}$ ✓

5.4.3 (3)

$= \dfrac{4}{5} \cdot \dfrac{\sqrt{2}}{2} - \dfrac{3}{5} \cdot \dfrac{\sqrt{2}}{2}$ ✓✓$= \dfrac{\sqrt{2}}{10}$ ✓

5.5.1 (6)

$= \dfrac{(-\sin x)(\sin x)}{(-\sin x)(-\tan x)}$ ✓✓✓✓

$= -\dfrac{\sin x}{\tan x} = -\sin x \cdot \dfrac{\cos x}{\sin x}$ ✓

$= -\cos x$ ✓

5.5.2 (5)

$\sqrt{\sin^2 2A} = \sin 2A$ (positive, since $0^\circ < 2A < 180^\circ$) ✓✓

denominator $= \cos A(-\sin A)$ ✓

$= \dfrac{2\sin A \cos A}{-\sin A \cos A}$ ✓

$= -2$ ✓

5.5.3 (5)

$= \dfrac{\sin 2x \cos x - \cos 2x \sin x}{-\sin x}$ ✓✓

$= \dfrac{\sin(2x - x)}{-\sin x}$ ✓✓

$= -1$ ✓

5.5.4 (5)

$= \dfrac{\left(-\dfrac{1}{\sqrt{3}}\right)\left(-\dfrac{\sqrt{3}}{2}\right)}{\left(-\dfrac{\sqrt{3}}{2}\right)\left(\dfrac{\sqrt{3}}{2}\right)}$ ✓✓✓✓

$= \dfrac{\dfrac{1}{2}}{-\dfrac{3}{4}} = -\dfrac{2}{3}$ ✓

5.5.5 (4)

$$= \frac{2(2\sin 15^\circ \cos 15^\circ)}{\cos 150^\circ} \checkmark\checkmark$$
$$= \frac{2\sin 30^\circ}{-\cos 30^\circ} = \frac{1}{-\frac{\sqrt{3}}{2}} \checkmark$$
$$= -\frac{2}{\sqrt{3}} \checkmark$$

5.5.6 (5)

$\cos 70^\circ = \sin 20^\circ, \cos 69^\circ = \sin 21^\circ, \ldots$ ✓

Pair the terms: $(\cos^2 20^\circ + \sin^2 20^\circ) + \ldots + (\cos^2 44^\circ + \sin^2 44^\circ) + \cos^2 45^\circ$ ✓

$= 25 \times 1$ ✓$+ \left(\frac{1}{\sqrt{2}}\right)^2$ ✓$= 25\frac{1}{2}$ ✓

5.5.7 (5)

$$\left(x - \frac{1}{x}\right)^2 = x^2 - 2 + \frac{1}{x^2} = 3 - 2 = 1 \checkmark\checkmark$$
$$4\sin^2 A = 1 \Rightarrow \sin^2 A = \frac{1}{4} \checkmark$$
$$\cos 2A = 1 - 2\sin^2 A \checkmark = \frac{1}{2} \checkmark$$

5.5.8 (4)

$= 2\sin(\alpha + 90^\circ)$ [double angle] ✓✓

$= 2\cos\alpha$ ✓

$= 2\sqrt{1 - p^2}$ ✓

5.6.1 (3)

$\cos(\alpha + \beta) = \cos(\alpha - (-\beta))$ ✓

$= \cos\alpha\cos(-\beta) + \sin\alpha\sin(-\beta)$ ✓

$= \cos\alpha\cos\beta - \sin\alpha\sin\beta$ ✓

5.6.2 (3)

$\sin(\alpha + \beta) = \cos[90^\circ - (\alpha + \beta)] = \cos[(90^\circ - \alpha) - \beta]$ ✓

$= \cos(90^\circ - \alpha)\cos\beta + \sin(90^\circ - \alpha)\sin\beta$ ✓

$= \sin\alpha\cos\beta + \cos\alpha\sin\beta$ ✓

For $\sin(\alpha - \beta)$: $\cos[(90^\circ - \alpha) + \beta]$, expand with $\cos(\alpha + \beta)$ from 5.6.1.

5.6.3 (4)

LHS $= \sin A\cos B + \cos A\sin B - \sin A\cos B + \cos A\sin B = 2\cos A\sin B$ ✓✓

$\sin 75^\circ - \sin 15^\circ = \sin(45^\circ + 30^\circ) - \sin(45^\circ - 30^\circ) = 2\cos 45^\circ \sin 30^\circ$ ✓

$= 2 \cdot \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{2}}{2}$ ✓

5.7.1 (5)

$$\text{LHS} = \frac{2\sin x\cos x + \sin x}{2\cos^2 x - 1 + \cos x + 1} \checkmark\checkmark$$
$$= \frac{\sin x(2\cos x + 1)}{\cos x(2\cos x + 1)} \checkmark\checkmark$$
$$= \frac{\sin x}{\cos x} = \tan x = \text{RHS} \checkmark$$

5.7.2 (3)

$\cos x = 0$: $x = 90^\circ$ or 270° ✓

$2\cos x + 1 = 0$: $\cos x = -\frac{1}{2}$, $x = 120^\circ$ or 240° ✓✓

5.7.3 (4)

LHS $= (\cos x\cos y - \sin x\sin y)(\cos x\cos y + \sin x\sin y)$ ✓

$= \cos^2 x\cos^2 y - \sin^2 x\sin^2 y$ ✓

$= \cos^2 x(1 - \sin^2 y) - (1 - \cos^2 x)\sin^2 y$ ✓

$= \cos^2 x - \sin^2 y$ ✓

5.7.4 $= \cos(52{,}5^\circ + 7{,}5^\circ)\cos(52{,}5^\circ - 7{,}5^\circ) = \cos 60^\circ \cos 45^\circ$ ✓✓ (3)

$= \frac{1}{2} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{4}$ ✓

5.7.5 $\text{LHS} = \frac{(\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x)}{\cos x - \sin x}$ ✓ (4)

$= \frac{(\cos x - \sin x)(\cos x + \sin x)(1)}{\cos x - \sin x}$ ✓✓

$= \cos x + \sin x$ ✓

5.7.6 $\text{LHS} = \frac{(\cos x + \sin x)^2 - (\cos x - \sin x)^2}{(\cos x - \sin x)(\cos x + \sin x)}$ ✓ (5)

$= \frac{4 \sin x \cos x}{\cos^2 x - \sin^2 x}$ ✓✓

$= \frac{2 \sin 2x}{\cos 2x}$ ✓

$= 2 \tan 2x$ ✓

5.7.7 $= \frac{(\cos x - \sin x)(\cos^2 x + \sin x \cos x + \sin^2 x)}{\cos x - \sin x}$ ✓✓ (3)

$= 1 + \sin x \cos x$ ✓

5.7.8 Denominator: $-\cos x - 1 = -(\cos x + 1)$ ✓✓ (5)

Numerator: $(2\cos x - 1)(\cos x + 1)$ ✓

$= \frac{(2\cos x - 1)(\cos x + 1)}{-(\cos x + 1)}$ ✓

$= 1 - 2\cos x$ ✓

5.7.9 $\sin 3x = \sin(2x + x) = \sin 2x \cos x + \cos 2x \sin x$ ✓ (6)

$\sin x = \sin(2x - x) = \sin 2x \cos x - \cos 2x \sin x$ ✓

$\sin 3x - \sin x = 2\cos 2x \sin x$ ✓✓

$\frac{2\cos 2x \sin x}{2\cos 2x} = \sin x$ ✓✓

5.7.10 $= 8 \sin x \cos x(\cos^2 x - \sin^2 x) = 4 \sin 2x \cos 2x$ ✓✓ (5)

$= 2 \sin 4x$ ✓

Maximum 2 ✓, when $4x = 90^\circ$, $x = 22{,}5^\circ$ ✓

5.7.11 $= \frac{(-\sin x)(-\sin x)}{-\sin x}$ ✓ (5)

$= -\sin x$ ✓

Real if $-\sin x \geq 0$, i.e. $\sin x \leq 0$ ✓

and $\sin x \neq 0$ (denominator), so $x \in (180^\circ; 360^\circ)$ ✓✓

$\sin(540^\circ + x) = \sin(180^\circ + x) = -\sin x$. The 2025 memo gave $[180^\circ; 360^\circ]$; the end points make the denominator zero, so leave them out.

5.8.1 $6(1 - \sin^2 x) + 5 \sin x - 7 = 0 \Rightarrow 6 \sin^2 x - 5 \sin x + 1 = 0$ ✓✓ (6)

$(3 \sin x - 1)(2 \sin x - 1) = 0$ ✓

$\sin x = \frac{1}{3}$: $x = 19{,}47^\circ$ or $x = 160{,}53^\circ$ ✓

$\sin x = \frac{1}{2}$: $x = 30^\circ$ or $x = 150^\circ$ ✓✓

5.8.2 (6)

$\tan x = -3$ ✓: $x = 108{,}43^\circ + k \cdot 180^\circ$ ✓

$\cos 2x = \frac{1}{2}$ ✓: $2x = \pm 60^\circ + k \cdot 360^\circ$ ✓

$x = \pm 30^\circ + k \cdot 180^\circ, k \in \mathbb{Z}$ ✓✓

5.8.3 (6)

$\frac{\sin x}{\cos x} = 4 \sin x \Rightarrow \sin x(1 - 4\cos x) = 0$ ✓✓

$\sin x = 0$: with $\cos x > 0$, $x = k \cdot 360^\circ$ ✓✓

$\cos x = \frac{1}{4}$: $x = \pm 75{,}52^\circ + k \cdot 360^\circ, k \in \mathbb{Z}$ ✓✓

Never divide by $\sin x$: you lose the solutions of $\sin x = 0$. The condition $\cos x > 0$ rejects $x = 180^\circ + k \cdot 360^\circ$.

5.8.4 (6)

$\sin(50^\circ - x) = \cos 2x = \sin(90^\circ - 2x)$ ✓✓

$50^\circ - x = 90^\circ - 2x + k \cdot 360^\circ \Rightarrow x = 40^\circ + k \cdot 360^\circ$ ✓✓

or $50^\circ - x = 180^\circ - (90^\circ - 2x) + k \cdot 360^\circ \Rightarrow -3x = 40^\circ + k \cdot 360^\circ$ ✓

$x = -13{,}33^\circ + k \cdot 120^\circ, k \in \mathbb{Z}$ ✓

5.8.5 (7)

$2\sin^2 x = \cos^2 x + \sin x \cos x$ ✓

$2\sin^2 x - \sin x \cos x - \cos^2 x = 0 \Rightarrow (2\sin x + \cos x)(\sin x - \cos x) = 0$ ✓✓

$\tan x = -\frac{1}{2}$ or $\tan x = 1$ ✓

$\tan x = 1$ gives $\sin^2 x = \cos^2 x$, so $d = 0$: reject ✓

$x = 153{,}43^\circ + k \cdot 180^\circ, k \in \mathbb{Z}$ ✓✓

[175]

QUESTION 6

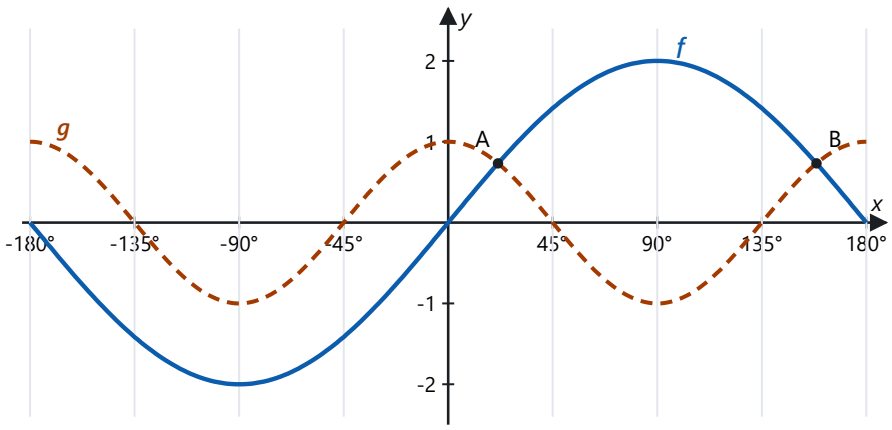
Trigonometry: Graphs

6.1 (4)

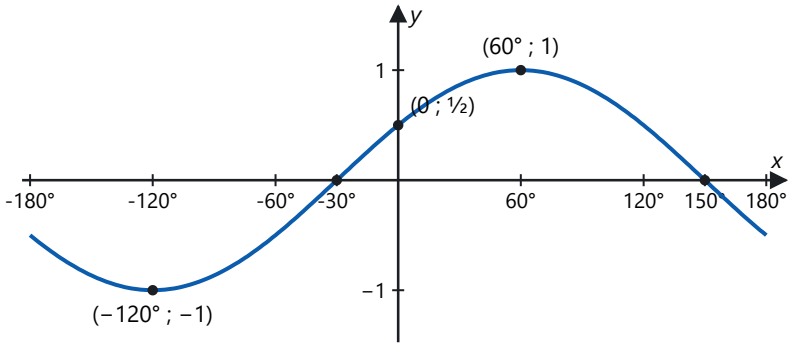
x-intercepts at $\pm 45^\circ$ and $\pm 135^\circ$ ✓

turning points $(-180^\circ; 1)$, $(-90^\circ; -1)$, $(0^\circ; 1)$, $(90^\circ; -1)$, $(180^\circ; 1)$ ✓✓

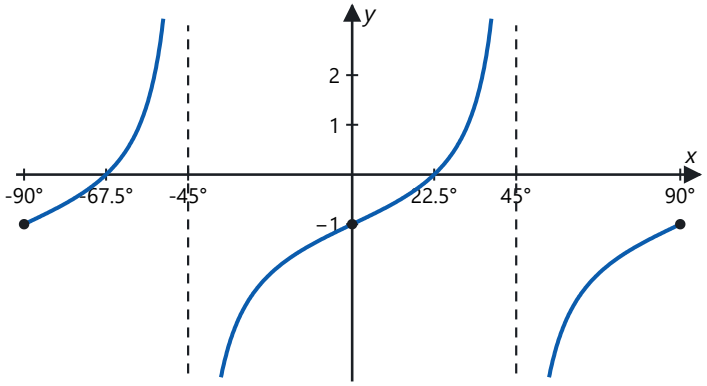
shape ✓



6.2 turning points $(60°; 1)$ and $(-120°; -1)$ ✓; intercepts $(-30°; 0)$, $(150°; 0)$, $\left(0; \frac{1}{2}\right)$ ✓; end points $\left(\pm 180°; -\frac{1}{2}\right)$ ✓ (3)



6.3 asymptotes $x = \pm 45°$ ✓; x-intercepts $-67{,}5°$ and $22{,}5°$ ✓; y-intercept -1 ✓; end points $(\pm 90°; -1)$ ✓ (4)



6.4 period of g: $180°$ ✓; amplitude of f: 2 ✓ (2)

Period of $f(3x) = 2\sin 3x$: $\frac{360°}{3} = 120°$.

6.5 $\frac{1}{2}f(x) = \sin x \in [-1; 1]$ ✓, so $h(x) \in [0; 2]$ ✓ (2)

Max of $f(x) - 3$: $2 - 3 = -1$.

6.6 $g(0°) = 1$ and $g(60°) = \cos 120° = -\frac{1}{2}$ ✓, so $-\frac{1}{2} \leq y \leq 1$ ✓ (2)

6.7 $2\sin x = 1 - 2\sin^2 x$ ✓ (4)

$2\sin^2 x + 2\sin x - 1 = 0$ ✓

$\sin x = \frac{-2 \pm \sqrt{4+8}}{4}$ ✓

$= \frac{-1 \pm \sqrt{3}}{2}$; $\frac{-1-\sqrt{3}}{2} < -1$ is not possible ✓

6.8 $x = 21{,}47°$ or $x = 180° - 21{,}47° = 158{,}53°$ ✓✓ (3)

$y = 2\sin x = 0{,}73$: A$(21{,}47°; 0{,}73)$, B$(158{,}53°; 0{,}73)$ ✓

6.9 (4)

$\sin x = \dfrac{-1+\sqrt{3}}{2}$ (from 6.7) ✓

$x = 21{,}47^\circ + k \cdot 360^\circ$ ✓

or $x = 158{,}53^\circ + k \cdot 360^\circ$ ✓

$k \in \mathbb{Z}$ ✓

6.10 (3)

$\tan 2x = \dfrac{1}{2}$ ✓; $2x = 26{,}57^\circ + k \cdot 180^\circ$ ✓; $x = 13{,}28^\circ + k \cdot 90^\circ$, $k \in \mathbb{Z}$ ✓

6.11 $21{,}47^\circ < x < 158{,}53^\circ$ ✓✓ (2)

6.12 $x \in [-135^\circ; -45^\circ]$ ✓$\cup [0^\circ; 45^\circ] \cup [135^\circ; 180^\circ]$ ✓and $x = -180^\circ$ ✓ (3)

Both negative, or both positive, or one of them zero. Check every end point.

6.13 $\sin x \leq -\dfrac{1}{2}$ ✓: $-150^\circ \leq x \leq -30^\circ$ ✓ (2)

6.14 $x \in [-180^\circ; -90^\circ]$ ✓ or $x \in [90^\circ; 180^\circ]$ ✓ (2)

6.15 Shift the solutions 10° left: $x = 11{,}47^\circ$ ✓ or $x = 148{,}53^\circ$ ✓ (2)

6.16 $p(x) = \cos 2(x + 45^\circ) = \cos(2x + 90^\circ)$ ✓$= -\sin 2x$ ✓ (2)

6.17 (3)

$-2\sin k = 1 \Rightarrow \sin k = -\dfrac{1}{2}$ ✓

$k = -30^\circ$ ✓ or $k = -150^\circ$ ✓

6.18 $t = -1$ ✓✓ (2)

$t = 1$ gives three solutions (-180°, 0°, 180°); $-1 < t < 1$ gives four.

6.19 $PQ = g(x) - f(x) = 1 - 2\sin^2 x - 2\sin x$ ✓✓ (6)

Maximum where $\sin x = -\dfrac{b}{2a} = -\dfrac{-2}{2(-2)} = -\dfrac{1}{2}$ ✓

$PQ_{\max} = 1 - 2\left(\dfrac{1}{4}\right) + 1 = \dfrac{3}{2}$ ✓

$x = -30^\circ$ or $x = -150^\circ$ ✓✓

6.20 $2\sin(x + 30^\circ) = 2(\sin x \cos 30^\circ + \cos x \sin 30^\circ) = \sqrt{3}\sin x + \cos x$ ✓ (4)

$\sin(x + 30^\circ) \geq \dfrac{1}{2}$ ✓

$30^\circ \leq x + 30^\circ \leq 150^\circ$ ✓

$0^\circ \leq x \leq 120^\circ$ ✓

6.21 period 180° ✓; $x = -45^\circ$ and $x = 135^\circ$ ✓ (2)

6.22 (3)

$\tan(t - 45^\circ) = 1$ ✓

$t - 45^\circ = 45^\circ + k \cdot 180^\circ$ ✓

$t = 90^\circ + k \cdot 180^\circ$, $k \in \mathbb{Z}$ ✓

6.23 $-90^\circ \leq x < -45^\circ$ ✓ or $90^\circ \leq x < 135^\circ$ ✓ (3)

Round brackets at the asymptotes: h is undefined there.

✓

[67]

QUESTION 7

Trigonometry: 2D And 3D Problems

7.1.1 90° (PQ is vertical, so it is $\perp$ to every line in the horizontal plane) ✓ (1)

7.1.2 $\tan\theta = \dfrac{h}{QS}$ ✓$\Rightarrow QS = \dfrac{h}{\tan\theta}$ ✓ (2)

7.1.3 $RS^2 = QR^2 + QS^2 - 2(QR)(QS)\cos 2\theta$ [cosine rule] ✓ (5)

$= 2QS^2(1 - \cos 2\theta) = 2QS^2(2\sin^2\theta)$ ✓✓

$RS = 2QS\sin\theta = 2 \cdot \dfrac{h\cos\theta}{\sin\theta} \cdot \sin\theta$ ✓

$= 2h\cos\theta$ ✓

7.1.4 $RS = 2(12)\cos 35^\circ$ ✓$= 19{,}66$ m ✓ (2)

7.1.5 $QS = \dfrac{12}{\tan 35^\circ} = 17{,}14$ ✓ (3)

Area $= \dfrac{1}{2}(QS)(QR)\sin 2\theta = \dfrac{1}{2}(17{,}14)^2\sin 70^\circ$ ✓

$= 138{,}00$ m² ✓

7.1.6 $\dfrac{1}{2}(15)(15)\sin R\hat{Q}S = 100$ ✓ (3)

$\sin R\hat{Q}S = \dfrac{200}{225} = 0{,}8889$ ✓

$\text{R}\hat{\text{Q}}\text{S} = 62{,}73^\circ$ ✓

7.2.1 $\hat{A} = 180^\circ - (60^\circ + x) = 120^\circ - x$ ✓ (5)

$\dfrac{CD}{\sin(120^\circ - x)} = \dfrac{3p}{\sin x}$ [sine rule] ✓

$\sin(120^\circ - x) = \sin 120^\circ\cos x - \cos 120^\circ\sin x = \dfrac{\sqrt{3}}{2}\cos x + \dfrac{1}{2}\sin x$ ✓✓

$CD = \dfrac{3p\left(\dfrac{\sqrt{3}}{2}\cos x + \dfrac{1}{2}\sin x\right)}{\sin x} = \dfrac{3p(\sqrt{3}\cos x + \sin x)}{2\sin x}$ ✓

7.2.2 $CD = 14{,}72$ ✓ (3)

Area $= \dfrac{1}{2}(AD)(CD)\sin 60^\circ = \dfrac{1}{2}(12)(14{,}72)\sin 60^\circ$ ✓

$= 76{,}49$ square units ✓

7.3.1 $DB^2 = 6^2 + 6^2 = 72$ [Pythagoras] ✓; $OB = \dfrac{1}{2}\sqrt{72}$ ✓$= 3\sqrt{2} = 4{,}24$ m ✓ (3)

7.3.2 $EB^2 = EO^2 + OB^2 = x^2 + 18$ ✓ (4)

EA = EB, so the perpendicular from E to AB bisects AB: the foot is 3 m from B ✓

$\cos\theta = \dfrac{3}{EB}$ ✓$= \dfrac{3}{\sqrt{x^2 + 18}}$ ✓

The 2016 memo used the cosine rule in △EAB instead; both earn full marks.

7.3.3 $\dfrac{1}{3}(36)x = 48$ ✓$\Rightarrow x = 4$ ✓ (4)

$\cos\theta = \dfrac{3}{\sqrt{16 + 18}} = \dfrac{3}{\sqrt{34}}$ ✓

$\theta = 59{,}04^\circ$ ✓

7.4.1 $\dfrac{1}{2}(BC)(BD)\sin 30^\circ = 40$ ✓ (3)

$\dfrac{1}{2}(10)(BD)\left(\dfrac{1}{2}\right) = 40$ ✓

$BD = 16$ m ✓

7.4.2 $\hat{ADB} = \beta = 25^\circ$ [alt ∠s; angle of depression] ✓ (3)

$AB = 16\tan 25^\circ$ ✓

$= 7{,}46$ m ✓

7.4.3 $CD^2 = 10^2 + 16^2 - 2(10)(16)\cos 30^\circ$ ✓✓ (3)

$CD = 8{,}88$ m ✓

7.4.4 $12^2 = 10^2 + BE^2 - 2(10)(BE)\cos 30^\circ$ ✓ (4)

$BE^2 - 17{,}32\,BE - 44 = 0$ ✓

$BE = \dfrac{17{,}32 + \sqrt{17{,}32^2 + 176}}{2}$ ✓

$= 19{,}57$ m (reject the negative root) ✓

7.5 The largest angle is opposite the longest side (12 m) ✓ (3)

$\cos\alpha = \dfrac{7^2 + 9^2 - 12^2}{2(7)(9)}$ ✓

$\alpha = 96{,}38^\circ$ ✓

[51]

QUESTION 8

Euclidean Geometry: Circles (Calculations And Theorem Proofs)

8.1.1 (2)

Statement	Reason
$\hat{D}_1 = 38^\circ$ ✓	tan chord theorem ✓

8.1.2 (2)

Statement	Reason
$A\hat{O}B = 2\hat{D}_1 = 76^\circ$ ✓	∠ at centre = 2 × ∠ at circumference ✓

8.1.3 (3)

Statement	Reason
OA = OB ✓	radii
$O\hat{A}B = O\hat{B}A$	∠s opp equal sides ✓
$O\hat{A}B = \dfrac{180^\circ - 76^\circ}{2} = 52^\circ$ ✓	sum of ∠s of △

8.1.4 (3)

Statement	Reason
$\hat{D}_2 = 70^\circ - 38^\circ = 32^\circ$ ✓	
$\hat{A}_2 = \hat{D}_2 = 32^\circ$ ✓	∠s in the same seg ✓

8.1.5 (2)

Statement	Reason
$C\hat{B}D = \hat{A}_3 = 50^\circ$ ✓	∠s in the same seg ✓

8.1.6 (2)

Statement	Reason
$A\hat{B}C = 180^\circ - 70^\circ = 110^\circ$ ✓	opp ∠s of cyclic quad ✓

8.1.7 (2)

Statement	Reason
$D\hat{C}E = B\hat{A}D = 32^\circ + 50^\circ = 82^\circ$ ✓	ext ∠ of cyclic quad ✓

8.1.8 (3)

Statement	Reason
$\text{A}\hat{\text{B}}\text{D} = 180° - (82° + 38°) = 60°$ ✓	sum of ∠s of △
$\hat{A}_1 = 38° \neq \text{A}\hat{\text{B}}\text{D}$ ✓	
No, TS is not parallel to BD	alternate ∠s are not equal ✓

8.2 (5)

Statement	Reason
$\hat{A} = \frac{1}{2}\hat{O}_1 = \frac{3x + 40°}{2}$	∠ at centre = 2 × ∠ at circumference ✓ S/R
$\hat{A} + \hat{C} = 180°$ ✓	opp ∠s of cyclic quad ✓
$\frac{3x + 40°}{2} + 2x + 20° = 180° \Rightarrow 7x = 280°$ ✓	

$x = 40°$ ✓

Check: $\hat{O}_1 = 160°$, $\hat{A} = 80°$, $\hat{C} = 100°$.

8.3.1 (2)

Statement	Reason
$\hat{R} = 180° - 110° = 70°$ ✓	co-int ∠s; QW ∥ RK ✓

8.3.2 (2)

Statement	Reason
$\hat{P} = 180° - 70° = 110°$ ✓	opp ∠s of cyclic quad ✓

8.3.3 (3)

Statement	Reason
$\text{P}\hat{\text{Q}}\text{R} = \text{P}\hat{\text{S}}\text{T} = 128°$ ✓	ext ∠ of cyclic quad ✓
$\text{P}\hat{\text{Q}}\text{W} = 128° - 110° = 18°$ ✓	

8.3.4 (2)

Statement	Reason
$\text{P}\hat{\text{U}}\text{Q} = 180° - 110° - 18° = 52°$ ✓	sum of ∠s of △ ✓

8.4.1 Line from centre ⊥ chord bisects the chord (AM = ½AB) ✓ (1)

8.4.2 (3)

Statement	Reason
$OA^2 = OM^2 + AM^2 = 5^2 + 12^2$ ✓	Pythagoras ✓
$OA = 13$ units ✓	

8.4.3 (4)

Statement	Reason
CK = 5	line from centre ⊥ chord ✓ S/R
$OK^2 = 13^2 - 5^2 = 144$, so $OK = 12$ ✓	Pythagoras ✓
distance $= OK + OM = 12 + 5 = 17$ units ✓	

8.5 (5)

Statement	Reason
AT = 15 and AS = 24	line from centre ⊥ chord ✓ S/R
$OA^2 = OT^2 + 15^2 = OS^2 + 24^2$ ✓	Pythagoras ✓
$OT^2 - \frac{49}{400}OT^2 = 351 \Rightarrow OT^2 = 400$, $OT = 20$ ✓	

$OA = \sqrt{400 + 225} = 25$ units ✓

8.6.1 equal chords subtend equal ∠s ✓ (1)

8.6.2 (7)

Statement	Reason
reflex $\text{T}\hat{\text{O}}\text{P} = 360° - x$ ✓	∠s around a pt ✓
$\text{T}\hat{\text{M}}\text{P} = \frac{1}{2}(360° - x) = 180° - \frac{1}{2}x$ ✓	∠ at centre = 2 × ∠ at circumference ✓
$\text{M}\hat{\text{T}}\text{P}$ = $\text{M}\hat{\text{P}}\text{T}$	∠s opp equal sides ✓ S/R
$\text{M}\hat{\text{P}}\text{T} = \frac{1}{2}\left(\frac{1}{2}x\right) = \frac{1}{4}x$ ✓	sum of ∠s of △
$\text{S}\hat{\text{T}}\text{M}$ = $\text{M}\hat{\text{P}}\text{T} = \frac{1}{4}x$	tan chord theorem ✓

8.7.1 (5)

Statement	Reason
Construction: join OA and OB ✓	
OA = OB	radii ✓ S/R
AD = DB	given
OD = OD	common
△OAD ≡ △OBD	SSS ✓ S/R
$\hat{D}_1 = \hat{D}_2$	≡ △s
$\hat{D}_1 + \hat{D}_2 = 180°$	∠s on a str line ✓ S/R
$\hat{D}_1 = 90°$, so OD ⊥ AB ✓	

8.7.2 (5)

Statement	Reason
Construction: draw KO and produce it to M ✓	
$\hat{O}_1 = \hat{K}_1 + \hat{T}$	ext ∠ of △ ✓ S/R
$\hat{K}_1 = \hat{T}$	∠s opp equal sides; OK = OT (radii) ✓ S/R
$\hat{O}_1 = 2\hat{K}_1$ and similarly $\hat{O}_2 = 2\hat{K}_2$ ✓	
$\text{T}\hat{\text{O}}\text{P} = 2(\hat{K}_1 + \hat{K}_2) = 2\text{T}\hat{\text{K}}\text{P}$ ✓	

8.7.3 (5)

Statement	Reason
Construction: draw OK and OM ✓	
$\hat{O}_1 = 2\hat{J}$	∠ at centre = 2 × ∠ at circumference ✓ S/R
$\hat{O}_2 = 2\hat{L}$	∠ at centre = 2 × ∠ at circumference ✓ S/R
$\hat{O}_1 + \hat{O}_2 = 360°$	∠s around a point ✓ S/R
$2\hat{J} + 2\hat{L} = 360° \Rightarrow \hat{J} + \hat{L} = 180°$ ✓	

8.7.4 (5)

Statement	Reason
Construction: draw diameter NR and join RM ✓	
$\text{R}\hat{\text{N}}\text{Q} = 90°$	tan ⊥ radius ✓ S/R
$\text{N}\hat{\text{M}}\text{R} = 90°$	∠ in semicircle ✓ S/R
$\text{M}\hat{\text{N}}\text{Q} = 90° - \text{R}\hat{\text{N}}\text{M} = \hat{R}$	sum of ∠s of △ ✓ S/R
$\hat{R} = \hat{K}$, so $\text{M}\hat{\text{N}}\text{Q} = \hat{K}$	∠s in the same seg ✓ S/R

[74]

QUESTION 9

Euclidean Geometry: Circle Riders (Prove It)

9.1.1 tan chord theorem ✓ (1)

9.1.2 (5)

Statement	Reason
$\text{T}\hat{\text{E}}\text{C} = x + y$ ✓	ext ∠ of △ ✓
$\text{T}\hat{\text{C}}\text{E} = x + y$ ✓	
$\text{T}\hat{\text{E}}\text{C}$ = $\text{T}\hat{\text{C}}\text{E}$ ✓	
TC = TE	sides opp equal ∠s ✓

9.2.1 (6)

Statement	Reason
$\text{M}\hat{\text{B}}\text{A} = x$ ✓	∠s opp equal sides; MA = MB (tans from common pt) ✓
$\text{A}\hat{\text{C}}\text{B} = x$ ✓	tan chord theorem ✓
$\text{A}\hat{\text{D}}\text{M} = x$ ✓	corresp ∠s; DM ∥ CB ✓

9.2.2 (3)

Statement	Reason
$\text{A}\hat{\text{D}}\text{M}$ = $\text{A}\hat{\text{B}}\text{M} = x$ ✓	proved in 9.2.1
ADBM is a cyclic quad ✓	converse ∠s in the same seg ✓

9.2.3 (3)

Statement	Reason
$\text{M}\hat{\text{D}}\text{B}$ = $\text{M}\hat{\text{A}}\text{B} = x$ ✓	∠s in the same seg ✓
$\text{M}\hat{\text{D}}\text{B}$ = $\text{A}\hat{\text{D}}\text{M}$, so DM bisects $\text{A}\hat{\text{D}}\text{B}$ ✓	

9.2.4 (4)

Statement	Reason
$\text{D}\hat{\text{B}}\text{C}$ = $\text{M}\hat{\text{D}}\text{B} = x$ ✓	alt ∠s; DM ∥ CB ✓
$\hat{C} = x$ ✓	proved
DB = DC	sides opp equal ∠s ✓

9.3.1 (3)

Statement	Reason
$\hat{A}_1 + \hat{A}_2 = \hat{C}_1 = 100°$ ✓	ext ∠ of cyclic quad ✓
$\hat{A}_1 = 50°$ ✓	

9.3.2 (5)

Statement	Reason
$\hat{B} = 2\hat{A}_1 = 100°$ ✓	given
$\hat{D} = 80°$ ✓	opp ∠s of cyclic quad ✓
$\text{A}\hat{\text{C}}\text{D} = 180° - 80° - 50° = 50° = \hat{A}_2$ ✓	sum of ∠s of △
AD = DC	sides opp equal ∠s ✓

9.4.1 (3)

Statement	Reason
$\hat{R} = 180° - 90° - 25° = 65°$ ✓	sum of ∠s of △
$\text{Q}\hat{\text{T}}\text{S} = \hat{R} = 65°$ ✓	∠s in the same seg ✓

9.4.2 (4)

Statement	Reason
$\text{S}\hat{\text{P}}\text{Q} = 180° - 65° = 115°$ ✓	opp ∠s of cyclic quad ✓
$\hat{S}_1 = \hat{R} = 65°$ ✓	given
$\text{S}\hat{\text{P}}\text{Q} + \hat{S}_1 = 180°$, so PQ ∥ SR	co-int ∠s supplementary ✓

9.4.3 (3)

Statement	Reason
$\hat{Q}_1 = 90°$ ✓	co-int ∠s; PQ ∥ SR (and $\hat{QVR}$ = 90°) ✓
PT is a diameter	converse ∠ in semicircle ✓

9.5.1 (4)

Statement	Reason
$\hat{ONG} = 90°$ ✓	line from centre to midpoint of chord ✓
$\hat{OKG} = 90°$	tan ⊥ radius ✓ S/R
$\hat{ONG}$ + $\hat{OKG} = 180°$, so ONGK is a cyclic quad	converse opp ∠s of cyclic quad ✓

9.5.2 (2)

Statement	Reason
$\hat{KOG}$ = $\hat{KNG}$ ✓	∠s in the same seg (of circle ONGK) ✓

9.6.1 line from centre ⊥ chord bisects the chord ✓ (1)

9.6.2 ∠ in semicircle ✓ (1)

9.6.3 (5)

Statement	Reason
$\hat{PQT} = 90° - \hat{P}$ ✓	sum of ∠s of △PQT
$\hat{QRT} = 90° - \hat{P}$ ✓	sum of ∠s of △PQR ✓
$\hat{PQT}$ = $\hat{QRT}$ ✓	
PQ is a tangent to circle QTR	converse tan chord theorem ✓

9.7.1 (2)

Statement	Reason
$\hat{B}_1 = x$ ✓	∠s in the same seg ✓

9.7.2 (2)

Statement	Reason
$\hat{B}_2 = y$ ✓	ext ∠ of cyclic quad BDRT ✓

9.7.3 (3)

Statement	Reason
$\hat{C} = 180° - (x + y)$ ✓	sum of ∠s of △ACR
$\hat{SBD} + \hat{C} = (x + y) + 180° - (x + y) = 180°$ ✓	
SCDB is a cyclic quad	converse opp ∠s of cyclic quad ✓

9.8.1 (4)

Statement	Reason
$\hat{TAC}$ = $\hat{AEC}$ ✓	tan chord theorem (small circle) ✓
$\hat{TAB}$ = $\hat{ADB}$ ✓	tan chord theorem (large circle) ✓
$\hat{AEC}$ = $\hat{ADB}$, so CE ∥ BD	corresp ∠s equal

9.8.2 (2)

Statement	Reason
$\frac{AC}{AB} = \frac{AE}{AD}$ ✓	line ∥ one side of △; CE ∥ BD ✓

9.9 If SD were a diameter, $\hat{SBD}$ would be $90°$ ✓ [∠ in semicircle] ✓. But $\hat{SBD} = 110° \neq 90°$ ✓, so SD is not a diameter. ✓ (4)

[70]

10.1 (6)

Statement	Reason
Construction: join DC and BE; draw heights h from E ⊥ AD and k from D ⊥ AE ✓	
$\frac{\text{area } \triangle ADE}{\text{area } \triangle DBE} = \frac{\frac{1}{2}AD \cdot h}{\frac{1}{2}DB \cdot h} = \frac{AD}{DB}$ ✓	same height
$\frac{\text{area } \triangle ADE}{\text{area } \triangle EDC} = \frac{\frac{1}{2}AE \cdot k}{\frac{1}{2}EC \cdot k} = \frac{AE}{EC}$ ✓	same height
area △DBE = area △EDC ✓	same base DE; same height; DE ∥ BC ✓
$\frac{AD}{DB} = \frac{AE}{EC}$ ✓	

10.2 (6)

Statement	Reason
Construction: mark P on AB and Q on AC with AP = DE and AQ = DF; join PQ ✓	
△APQ ≡ △DEF ✓	SAS ✓
$\text{A}\hat{\text{P}}\text{Q} = \hat{E} = \hat{B}$ ✓	
PQ ∥ BC	corresp ∠s equal ✓ S/R
$\frac{AB}{AP} = \frac{AC}{AQ} \Rightarrow \frac{AB}{DE} = \frac{AC}{DF}$	line ∥ one side of △ ✓ S/R

10.3.1 (2)

Statement	Reason
$\frac{PT}{PR} = \frac{PS}{PQ} = \frac{3}{5}$, so PT = 18 ✓	line ∥ one side of △ ✓

10.3.2 (2)

Statement	Reason
RU : UQ = RT : TP = 2 : 3 ✓	line ∥ one side of △; TU ∥ PQ ✓

10.3.3 (3)

Statement	Reason
$\hat{P}$ common; $\text{P}\hat{\text{S}}\text{T} = \hat{Q}$ ✓	corresp ∠s; ST ∥ QR
△PST ||| △PQR	∠∠∠ ✓
$\frac{ST}{QR} = \frac{PS}{PQ} = \frac{3}{5} \Rightarrow ST = 12$ ✓	

10.3.4 (3)

$$= \frac{\frac{1}{2}(PS)(PT)\sin P}{\frac{1}{2}(PQ)(PR)\sin P} \checkmark$$

$$= \frac{3}{5} \times \frac{3}{5} \checkmark$$

$$= \frac{9}{25} \checkmark$$

10.3.5 (4)

△TUR ||| △PQR with ratio $\frac{TR}{PR} = \frac{2}{5}$, so area △TUR $= \frac{4}{25}$ of △PQR ✓✓

SQUT $= 1 - \frac{9}{25} - \frac{4}{25}$ ✓

$= \frac{12}{25}$ ✓

10.4.1 (4)

Statement	Reason
$\frac{AE}{EC} = \frac{AD}{DB}$ ✓	line ‖ one side of △ ✓
$\frac{x+1}{2x-2} = \frac{3}{4} \Rightarrow 4x + 4 = 6x - 6$ ✓	
$x = 5$ ✓	

10.4.2 Midpoint theorem ✓; DE = ½BC ✓ (2)

10.5.1 (5)

Statement	Reason
$\frac{AC}{CH} = \frac{AF}{FG} \Rightarrow \frac{AC}{14} = \frac{12}{8}$ ✓	line ‖ one side of △ ✓
AC = 21 ✓	
BD = AC = 21 and DM $= \frac{1}{2}$BD $= 10{,}5$ ✓	diagonals of a rectangle are equal and bisect each other ✓

10.6.1 (2)

Statement	Reason
$\frac{AF}{FB} = \frac{4}{3} = \frac{AG}{GC}$ ✓	
FG ‖ BC	converse: line divides two sides of △ in proportion ✓

10.6.2 (3)

Statement	Reason
$\frac{AH}{HK} = \frac{AG}{GC}$ ✓	line ‖ one side of △; GH ‖ CK ✓
$\frac{AE}{ED} = \frac{AG}{GC} = \frac{AH}{HK}$ ✓	line ‖ one side of △; GE ‖ CD

10.6.3 (4)

$\frac{8}{HK} = \frac{4}{3} \Rightarrow HK = 6$ ✓

$\frac{AE}{9} = \frac{4}{3} \Rightarrow AE = 12$ ✓

AK = 8 + 6 = 14 ✓

EK = 14 − 12 = 2 ✓

10.7.1 (3)

Statement	Reason
$\frac{GL}{LM} = \frac{FK}{KM} = 3$ ✓	line ‖ one side of △; GF ‖ LK ✓
GL = 3y and LH = HG, so GH $= \frac{3}{2}y$ ✓	

10.7.2 (5)

Statement	Reason
$\hat{M}$ common ✓	common
$\hat{MHF}$ = $\hat{LKM}$ ✓	ext ∠ of cyclic quad ✓
$\hat{LKM}$ = $\hat{GFM}$	corresp ∠s; GF ‖ LK ✓ S/R
△MFH ||| △MGF	∠∠∠ ✓

10.7.3 (3)

$\frac{MF}{MG} = \frac{MH}{MF} \Rightarrow MF^2 = MH \cdot MG$ ✓

$(4x)^2 = \left(\frac{5}{2}y\right)(4y)$ ✓

$16x^2 = 10y^2 \Rightarrow \frac{y}{x} = \sqrt{\frac{8}{5}}$ ✓

10.8.1 (2)

Statement	Reason
$\hat{A} = x$ ✓	tan chord theorem ✓

10.8.2 (3)

Statement	Reason
$\hat{D}$ is common ✓	common
$\text{D}\hat{\text{C}}\text{B} = \hat{A} = x$ ✓	proved
△DCB ||| △DAC	∠∠∠ ✓

10.8.3 (2)

Statement	Reason
$\frac{DC}{DA} = \frac{DB}{DC}$ ✓	||| △s ✓

10.8.4 (3)

DA = 4 + 12 = 16 ✓

$DC^2 = 4 \times 16 = 64$ ✓

DC = 8 units ✓

10.8.5 (3)

$\text{A}\hat{\text{C}}\text{B} = 90°$ [∠ in semicircle] ✓

$\frac{CB}{CA} = \frac{DC}{DA} = \frac{8}{16}$ [||| △s] ✓

In right-angled △ACB: $\tan \hat{A} = \frac{CB}{CA} = \frac{1}{2}$ ✓

10.8.6 (4)

Statement	Reason
$\hat{B}$ common; $\text{C}\hat{\text{E}}\text{B} = \text{A}\hat{\text{C}}\text{B} = 90°$ ✓	given; ∠ in semicircle
△CEB ||| △ACB	∠∠∠ ✓
$\frac{CB}{AB} = \frac{BE}{CB} \Rightarrow CB^2 = BE \cdot BA$ ✓	||| △s ✓

10.8.7 (4)

$CB^2 + CA^2 = 144$ and $CA = 2CB$ ✓

$5CB^2 = 144$, $CB^2 = 28{,}8$ ✓

$BE = \frac{CB^2}{AB} = \frac{28{,}8}{12} = 2{,}4$ ✓

$CE = \sqrt{28{,}8 - 2{,}4^2} = 4{,}8$ units ✓

10.8.8 (3)

$DC^2 = DB \cdot DA = DB(DB + BA)$ ✓

$= DB^2 + DB \cdot BA$ ✓

$DC^2 - DB^2 = DB \cdot BA$ ✓

10.8.9 (3)

$\hat{A} = x$ and $\text{A}\hat{\text{C}}\text{B} = 90°$, so $\sin x = \frac{CB}{AB}$ ✓

$CB^2 = BE \cdot BA$ (10.8.6) ✓

$\frac{BE}{BA} = \frac{CB^2}{AB^2} = \sin^2 x$ ✓

10.9.1 (4)

Statement	Reason
$\frac{EF}{DC} = \frac{6}{15} = \frac{2}{5}; \frac{FT}{CT} = \frac{4}{10} = \frac{2}{5}; \frac{ET}{DT} = \frac{5}{12,5} = \frac{2}{5}$ ✓	
All three ratios are equal ✓	
△EFT ||| △DCT ✓	sides of △ in proportion ✓

10.9.2 (3)

Statement	Reason
$\text{E}\hat{\text{F}}\text{D} = \text{E}\hat{\text{C}}\text{D}$ ✓	||| △s
EFCD is a cyclic quad	converse ∠s in the same seg ✓
$\text{D}\hat{\text{F}}\text{C} = \text{D}\hat{\text{E}}\text{C}$ ✓	∠s in the same seg

[91]